

Basics 13: Lorentz Force and Charged-Particle Motion

II - Electromagnetism foundations

Gyromotion, helical trajectories, electric acceleration, drift motion, and numerical integration

Textbook draft, version 1.0. The Lorentz equation is the microscopic starting point for orbit following, guiding-center theory, and kinetic descriptions. This chapter solves the simplest field configurations exactly and uses them to develop gyrofrequency, Larmor radius, pitch angle, and the electric-cross-magnetic drift. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Solve charged-particle motion in uniform electric and magnetic fields.
- Derive gyrofrequency and Larmor radius.
- Separate parallel and perpendicular motion.
- Derive the electric-cross-magnetic drift.
- Explain why drift velocity is independent of particle mass and charge sign.
- Assess numerical orbit integration using invariants.

2 Prerequisites and reading strategy

- Basics 2, 4, and 11.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Equation of motion and energy balance

For a nonrelativistic particle,

$$m \frac{d\mathbf{v}}{dt} = q (\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad \frac{d\mathbf{x}}{dt} = \mathbf{v}. \quad (3.1)$$

Dotting with $\mathbf{v}$ gives

$$\frac{d}{dt} \left(\frac{mv^2}{2} \right) = q\mathbf{v} \cdot \mathbf{E}. \quad (3.2)$$

Electric fields can change kinetic energy. Magnetic fields change direction but not speed. This separation is a powerful diagnostic for orbit codes.

4 Uniform magnetic field: gyromotion

Let $\mathbf{B} = B_0 \hat{\mathbf{z}}$ and $\mathbf{E} = 0$. Then

$$\dot{v}_x = \Omega_s v_y, \quad (4.1)$$

$$\dot{v}_y = -\Omega_s v_x, \quad (4.2)$$

$$\dot{v}_z = 0, \quad (4.3)$$

where the signed cyclotron frequency is

$$\Omega_s = \frac{qB_0}{m}. \quad (4.4)$$

Differentiating once more gives $\ddot{v}_x + \Omega_s^2 v_x = 0$. The perpendicular velocity rotates at $|\Omega_s|$, while $v_{\parallel} = v_z$ is constant. The Larmor radius is

$$\rho_L = \frac{v_{\perp}}{|\Omega_s|} = \frac{mv_{\perp}}{|q|B_0}. \quad (4.5)$$

The sign of q determines the direction of rotation.

5 Helical trajectories and pitch angle

Integrating velocity produces a circle in the plane perpendicular to $\mathbf{B}$ and uniform translation along $\mathbf{B}$. The result is a helix. Define pitch angle α by

$$v_{\parallel} = v \cos \alpha, \quad v_{\perp} = v \sin \alpha. \quad (5.1)$$

The distance advanced along the field during one gyroperiod $T_c = 2\pi/|\Omega_s|$ is

$$L_{\text{pitch}} = v_{\parallel} T_c. \quad (5.2)$$

Do not confuse this geometric helix pitch with pitch variables used for magnetic-mirror or guiding-center coordinates.

6 Uniform electric field

With $\mathbf{B} = 0$ and constant $\mathbf{E}$,

$$\mathbf{v}(t) = \mathbf{v}_0 + \frac{q\mathbf{E}}{m}t, \quad \mathbf{x}(t) = \mathbf{x}_0 + \mathbf{v}_0 t + \frac{q\mathbf{E}}{2m}t^2. \quad (6.1)$$

For $\mathbf{E} = E_{\parallel} \hat{\mathbf{b}}$ in the presence of a uniform magnetic field, the parallel motion accelerates while perpendicular gyromotion persists. A sustained parallel electric field therefore changes both energy and pitch angle.

7 Crossed fields and the $\mathbf{E} \times \mathbf{B}$ drift

Let $\mathbf{E} \perp \mathbf{B}$ be constant. Seek a constant velocity $\mathbf{v}_E$ for which the transverse Lorentz force vanishes:

$$\mathbf{E} + \mathbf{v}_E \times \mathbf{B} = 0. \quad (7.1)$$

Cross with $\mathbf{B}$ and use $\mathbf{E} \cdot \mathbf{B} = 0$:

$$\boxed{\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2}}. \quad (7.2)$$

Writing $\mathbf{v} = \mathbf{v}_E + \mathbf{w}$ reduces the equation for $\mathbf{w}$ to ordinary gyromotion. The guiding center drifts at $\mathbf{v}_E$, independent of m , $|q|$, and charge sign. Hence ions and electrons share this bulk drift and it does not itself create a current in a uniform plasma.

8 Ordering toward guiding-center theory

If field scales L and times τ satisfy

$$\epsilon_{\rho} = \frac{\rho_L}{L} \ll 1, \quad \epsilon_{\Omega} = \frac{1}{|\Omega_s| \tau} \ll 1, \quad (8.1)$$

the rapid gyrophase can be separated from slow guiding-center motion. Nonuniform fields then produce grad- B , curvature, polarization, and other drifts. Those are developed in later kinetic notes.

9 Numerical orbit integration

A naive explicit Euler step can increase particle energy in a static magnetic field. Suitable checks include

$$\Delta K, \quad \Delta v_{\parallel}, \quad \Delta \rho_L, \quad \text{phase error per gyroperiod.} \quad (9.1)$$

The Boris algorithm is widely used because its magnetic rotation preserves speed to high accuracy. Adaptive Runge-Kutta methods may be accurate locally but do not automatically preserve long-time geometric structure.

10 Computational laboratory

Implement analytic and numerical trajectories for uniform $\mathbf{B}$, parallel $\mathbf{E}$, and crossed $\mathbf{E}$ and $\mathbf{B}$. Compare Euler, Runge-Kutta, and Boris updates over hundreds of gyroperiods. Measure energy, phase, guiding-center, and Larmor-radius errors.

11 Summary

- The Lorentz equation separates electric work from magnetic redirection.
- Uniform magnetic fields produce gyromotion characterized by gyrofrequency and Larmor radius.
- Parallel motion combines with gyromotion to form a helix.
- Crossed uniform fields produce a species-independent $\mathbf{E} \times \mathbf{B}$ drift.
- Orbit solvers should be judged by invariant and phase preservation, not only local truncation error.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
Ω_s	signed cyclotron frequency	qB/m .
ρ_L	Larmor radius	$v_\perp/ \Omega_s $.
$v_\parallel, v_\perp$	velocity components	Parallel and perpendicular to the field.
α	pitch angle	Angle between velocity and magnetic field.
$\mathbf{v}_E$	electric-cross-magnetic drift	$\mathbf{E} \times \mathbf{B}/B^2$.
T_c	gyroperiod	$2\pi/ \Omega_s $.

12 Exercises

1. Derive the gyromotion solution including the charge-sign dependence.
2. Show that the magnetic moment $mv_\perp^2/(2B_0)$ is constant in a uniform field.
3. Derive the $\mathbf{E} \times \mathbf{B}$ drift using a shifted velocity.

4. Calculate electron and deuteron gyrofrequencies at a specified magnetic field.
5. Explain why shared $\mathbf{E} \times \mathbf{B}$ motion produces no current in a uniform quasineutral plasma.

13 References

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